

BASIC TERMS

① Electric Current: Electric current may be defined as the time rate of net motion of an electric charge across a cross-sectional boundary.

i.e. electric current, I = Rate of transfer of electric charge.

$$I = \frac{\text{Quantity of electric charge transferred during a given time duration}}{\text{Time duration}} = \frac{dq}{dt}$$

Units

$$\text{Ampere} = \frac{\text{Coulomb}}{\text{seconds}}$$

i.e. Current is measured in "Ampere" or "Coulomb per second".

A random motion of electrons in a metal does not constitute a current unless there is a net transfer of charge with time.

② Electromotive force: (emf) It is the force that causes an electric current to flow in an electric circuit.

③ Potential difference: (pd) between two points in an electric circuit is that difference in their electrical state which tends to "cause" flow of electric current between them.

Note: Unit of both "emf" and "pd" is "Volt".
emf maintains potential difference while pd causes current to flow.

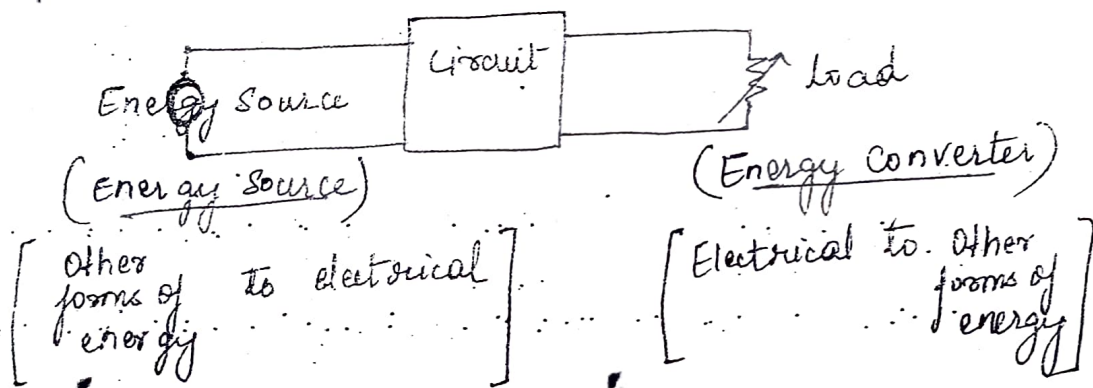
④ Electric Circuit: Electric circuit is closed energized network through which current flows.

⑤ Electric Network: It is any possible interconnection of electric circuit elements.

It is n/w but not circuit.

⑥ Energy Source : (or source), (such as a primary or secondary cell, a generator,) is a device that converts $\left\{ \begin{array}{l} \text{chemical,} \\ \text{mechanical,} \\ \text{thermal or} \\ \text{some other} \\ \text{forms of energy} \end{array} \right\}$ into $\left\{ \begin{array}{l} \text{electrical} \\ \text{Energy} \end{array} \right\}$.

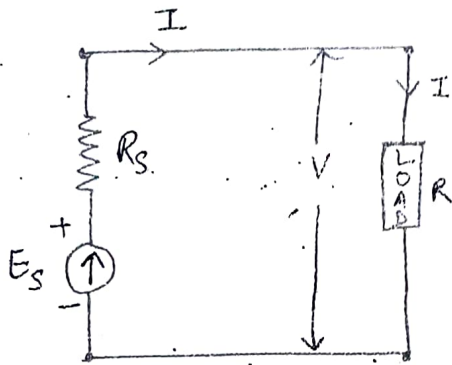
⑦ Energy Converter : It is also called the load, (such as lamp, heating appliance, or an electric motor) converts $\left(\begin{array}{l} \text{electrical} \\ \text{energy} \end{array} \right)$ into $\left(\begin{array}{l} \text{light,} \\ \text{heat,} \\ \text{mechanical work} \\ \text{and so on} \end{array} \right)$.



⑧ Source emf : (or open-circuit voltage) is that voltage that appears across the source when no load is connected across it.

⑨ Terminal Voltage : When a load is connected to the source terminals and the circuit is closed, an electric current starts flowing through the circuit. Now voltage across source terminals called terminal voltage. It is not equal to the source emf. It is due to the voltage drop inside the source, i.e. across the source resistance $= IR_s$.

⑧ Basic Circuit Diagram



→ A graphic representation of an electric circuit is called a circuit diagram.
 → Such a diagram consists of interconnected symbols called circuit elements or circuit parameters.

→ Two elements are necessary to represent processes in a dc circuit. These are emf source (E_s) and internal (source) resistance (R_s) and the load resistance (R).

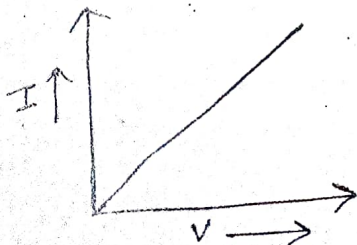
→ In energy converter (load), current flows from +ve side to -ve side of the source whereas inside the source current flows in the opposite direction i.e. from -ve side to the +ve side.

Various Types Of Circuits :

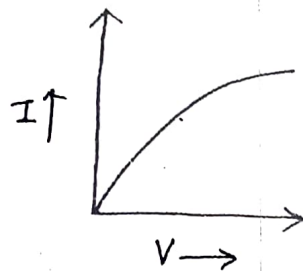
→ direct current (dc) circuit : When electrical energy is generated, transmitted and converted under conditions such that the current and voltages involved remain constant with time, the electric circuit is known as direct current (dc) circuit.

→ Alternating current (ac) circuit : If the currents and voltages do change with time, the circuit is known as alternating current (ac) circuit.

→ Linear circuit : Resistive elements for which the volt-ampere characteristics is a straight line i.e. they obey ohm's law are called linear and the electric circuits containing only linear resistances are called linear circuits.



→ Non-linear circuits: Resistive elements for which the volt-ampere characteristics is other than a straight line i.e. does not obey ohm's law, are known as ~~the~~ non-linear and so the electric circuits containing them are called non-linear circuits.



eg. tungsten lamps,
vacuum tubes,
transistors etc.

→ Bilateral circuit: An electric circuit, whose properties or characteristics are same in either direction, is called the bilateral circuit.

eg. distribution or transmission line because it can be made to perform its function equally well in either direction.

→ Unilateral circuit: An electric circuit, whose characteristics or properties change with the direction of its operation is called unilateral circuit.

eg. Diode rectifier because it cannot perform rectification in both directions.

CIRCUIT ELEMENTS : ACTIVE & PASSIVE & ENERGY STORAGE ASPECTS

Circuit elements or Circuit Parameters:

The various elements of an electric circuit are called its parameters like resistance, capacitance and inductance.

TWO TYPES

Active Elements
Active network: When a network contains one or more sources of emf and/or current, it is said to be active network.

* The elements which supply energy to the network are known as active elements.

Voltage sources like batteries, dc generators, ac generators or
current sources like photoelectric cells, magnetohydraulic generator, semiconductor devices like transistors etc.

Passive Elements:

Passive Network: A network is said to be passive if it contains no source of emf in it.

* The components which dissipate or store energy are known as passive components.

eg. Resistor (R), Inductor (L) & capacitor (C).

Resistor: It is the only component which dissipates electrical energy.

Inductor: It stores energy. Basically it stores energy by virtue of a current passing through it.

Capacitor: Stores energy. It stores energy by virtue of potential difference across it.

Resistance (R): It may be defined as the property of a substance due to which it opposes (or restricts) the flow of electric current through it.

Unit of resistance is ohm (Ω).

$$1 \text{ M}\Omega = 10^6 \Omega$$

$$1 \text{ K}\Omega = 10^3 \Omega$$

$$1 \text{ m}\Omega = 10^{-3} \Omega$$

$$1 \mu\Omega = 10^{-6} \Omega$$

The resistance 'R' offered by conductor depends on the following factors.

$$R = \rho \frac{l}{A}$$

- 'R' varies directly as its length (l) and varies inversely as the cross-sectional area of the conductor.
- It also depends on the nature of material and temperature of the conductors.

" ρ " is a constant depending on the nature of material of the conductor, and is known as its specific resistance or resistivity.

It may be defined as the resistance between the opposite faces of a metre cube of that material.

Units:

$$\begin{aligned} \rho &= \frac{R \cdot A}{l} \\ &= \frac{\text{ohm} \times \text{m}^2}{\text{m}} \end{aligned}$$

$$\rho = \text{ohm-metre}$$

$$\Omega\text{-m}$$

Conductance and Conductivity:

Conductance (G) is the reciprocal of resistance (R).

It is defined as the inducement offered by the conductor to the flow of current.

$$G = \frac{1}{R}$$

$$R = \frac{\rho l}{A}$$

$$G = \frac{1}{\frac{\rho l}{A}} = \frac{\sigma A}{l}$$

$$G = \frac{\sigma A}{l}$$

where " σ " is called the conductivity or specific conductance of a conductor.

Unit of conductance is siemens (S) or mho (Ω^{-1})

Unit of conductivity

$$\sigma = \frac{l G}{A} = \frac{\text{metre} \times (\text{siemens/mho})}{\text{metre}^2}$$

$$\sigma = \frac{\text{Siemens}}{\text{metre}} \text{ or } \frac{\text{ohm}^{-1} \text{m}}{\text{metre}}$$

EFFECT OF TEMPERATURE ON RESISTANCE:

The effect of rise in temperature is:

- (i) To increase the resistance of pure metals and alloys
- (ii) To decrease the resistance of electrolytes, insulators (paper, rubber, glass etc.) and partial conductors such as carbon. Hence, insulators possess -ve temperature coefficient of resistance.

TEMPERATURE COEFFICIENT OF RESISTANCE:

Let a Metallic conductor have

Resistance	R_0	R_t
temperature	0°C	$t^\circ\text{C}$

Increase in resistance $\Delta R = R_t - R_0$ depends

- (i) directly on its initial resistance.
- (ii) directly on the size in temperature.
- (iii) on the nature of the material of the conductor.

$$R_t - R_0 \propto R_0 \times t$$

$$R_t - R_0 = \alpha R_0 t$$

where α is constant and known as temperature coefficient of the resistance of the conductor.
It may be defined as the increase in resistance per ohm original resistance per $^\circ\text{C}$ rise in temperature.

Inductors: It opposes the change of current flowing through it and induces a voltage when the current flowing through it varies in magnitude and/or direction.

Capacitors: Voltage applied across the capacitor sets up an electric field within it and energy is stored in electric field.

Parameter	Basic Relationship	Voltage-Current Relationship	Energy
R	$V = Ri$	$V_R = Ri_R$ $i_R = \frac{1}{R} V_R = G V_R$	$W_R = \int_{-\infty}^t V_R i_R dt$
L	$\psi = Li$ (flux linkage)	$V_L = L \frac{di_L}{dt}$ $i_L = \frac{1}{L} \int_{-\infty}^t V_L dt$	$W_L = \frac{1}{2} LI^2$
C	$q = Cv$ (charge)	$i_C = C \frac{dv_C}{dt}$ $v_C = \frac{1}{C} \int_{-\infty}^t i_C dt$	$W_C = \frac{1}{2} CV^2$

Some Important Points Of Electric Network :

Node : It is a junction or point in ~~an~~ an electric network where two or more branches meet together.

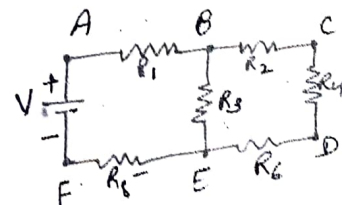
Branch : A branch is any section of a network which joins two nodes directly, i.e., without passing through a third node.

Loop : It is a closed path in a circuit in which no element or node is encountered more than once.

loop 1 : V, A, R₁, B, R₂, C, R₄, D, R₆, E, R₅, F

loop 2 : V, A, R₁, B, R₃, E, R₅, F

loop 3 : B, R₂, C, R₄, D, R₆, E, R₃.



Mesh : It is a loop that contains no other loop within it.

eg: V A R₁ B R₃ E R₅ F , B R₂ C R₄ D R₆ E R₃

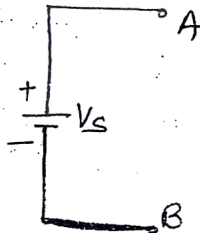
Independent & Dependent Sources (Current & Voltage)

VOLTAGE SOURCES & CURRENT SOURCES

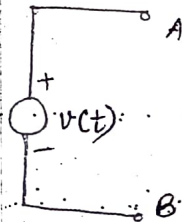
Independent Voltage Source:

A voltage source is said to be independent when it does not depend on any other quantity in the circuit.

Independent dc voltage source



Time-varying voltage source



+ve sign indicates

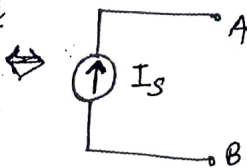
that terminal A is

positive with respect to terminal B i.e. the potential of terminal A is V volts higher than that of terminal B.

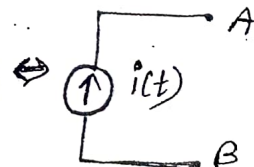
Independent Current Source:

A current source is said to be independent if it does not depend on any other quantity in the circuit.

constant current source



Time-varying current source



The arrow indicates the direction of flow of current at any moment under consideration.

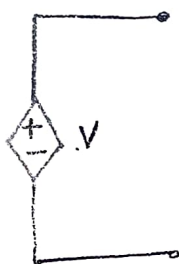
Dependent Sources

A dependent source is one which depends on some other quantity in the circuit which may be either a voltage or a current.

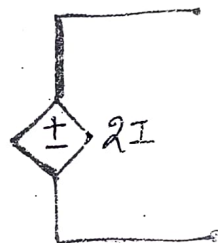
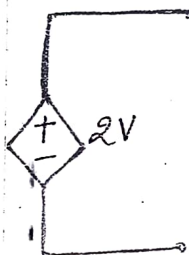
In a dependent source the output voltage (or current) depends on another voltage (or current).

The relationship may be linear or non-linear.

Dependent voltage source
is represented by



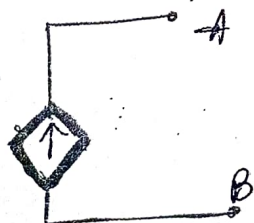
eg:
#



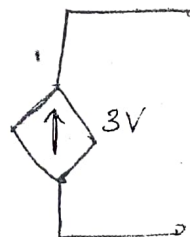
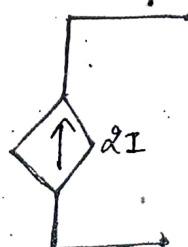
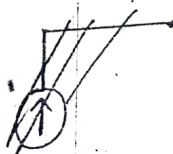
For finding out dependent voltage source, just check the (+) signs, then whatever be the ~~voltage~~ value is given either in terms of V or I , that will be the value of voltage only across the given dependent source.

eg: $2V$ & $2I$ both represents the voltage across dependent source.

Dependent current source
is represented by



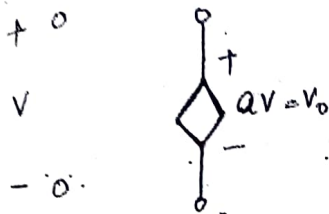
eg:
#



Here, arrow represents the current source.
Both $2I$ and $3V$ represents the value of current.

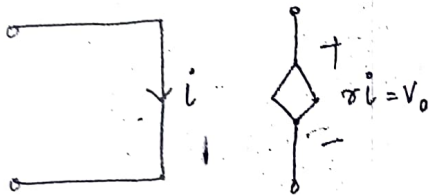
Four types of Dependent Sources

① Voltage-Dependent Voltage Source



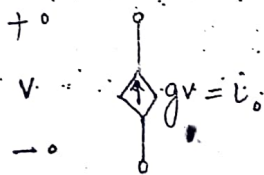
output voltage depends upon or is controlled by an input voltage in the ckt.

② Current-dependent Voltage Source



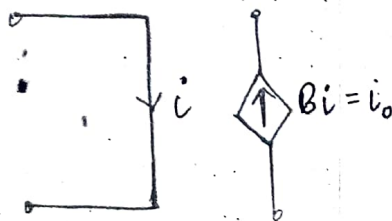
Output voltage depends upon current i.e. output voltage depends on or is controlled by an input current.

③ Voltage-dependent Current Source



Output current depends upon or is controlled by an input voltage.

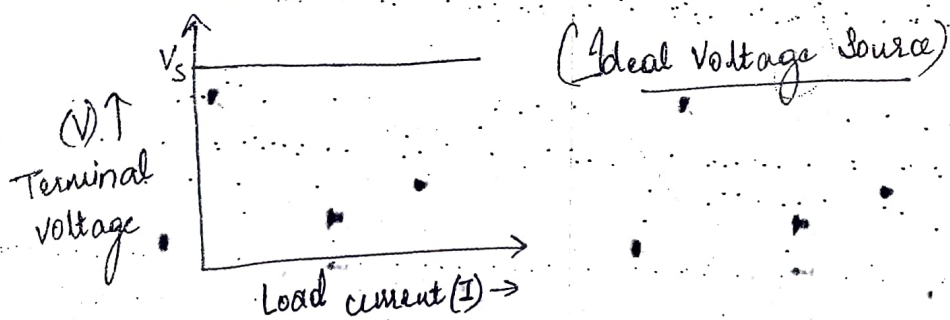
④ Current-Dependent Current Source



Output current depends upon or is controlled by an input current.

Ideal Voltage Source or Constant Voltage

- An ideal voltage source (also called constant voltage source) is one that maintains a constant terminal voltage, no matter how much current is drawn from it.
- An ideal voltage source has zero internal resistance.
 - ∴ it would provide constant terminal voltage regardless of the value of load connected across its terminals.
- It is not possible to construct an ideal voltage source because every voltage source has some internal resistance that causes the terminal voltage to fall due to the flow of current. So terminal voltage is not constant. However, if the internal resistance of voltage source is very small, it can be considered as a constant voltage source.
- Two important points:
 - (A) Ideal voltage source cannot be short-circuited.
 - (B) Two ideal voltage sources of unequal output voltages cannot be placed in parallel.



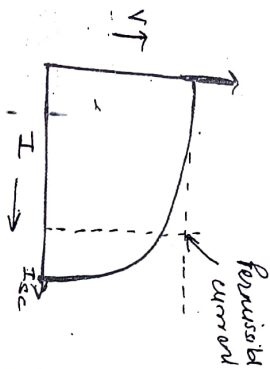
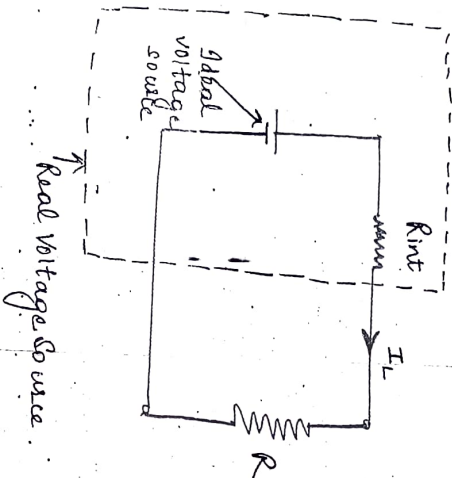
eg: Example of constant voltage source is lead-acid cell.

Internal resistance of this cell is very small (about 0.01Ω) so it can be considered as constant voltage source for all practical purposes.

Practical or Real Voltage Source:

A real (non-ideal) voltage source has internal resistance that causes its terminal voltage to decrease when load current is increased.

A real voltage source can be represented as an ideal voltage source in series with a resistance equal to its internal resistance (R_{int}).

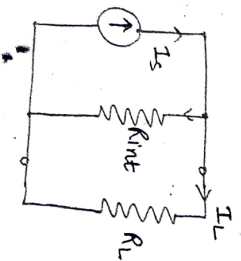


eg: generator or a battery.

Ideal Current Source or Constant Current Source.

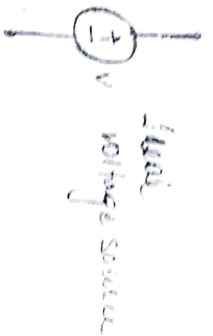
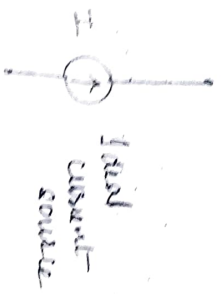
→ An ideal current source is one which will supply the same current to any resistance connected across its terminals.

→ An ideal current source has infinite resistance (internal).



In this fig. internal resistance (R_{int}) is infinite and may be considered as removed.
 ∴ current source will supply same current I_s to any resistance connected across its terminals.

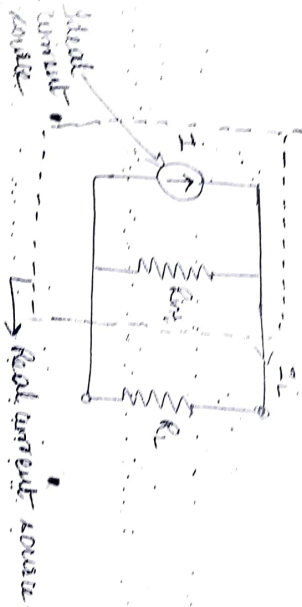
Key representation:



Ideal or Real Current Source:

→ Ideal current source does finite internal resistance i.e. the load current will change as the value of load resistance R_L changes.

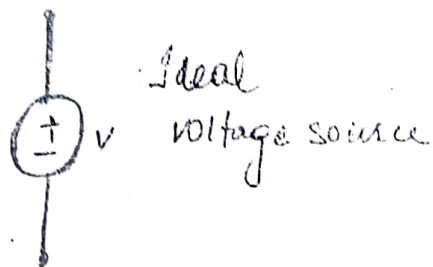
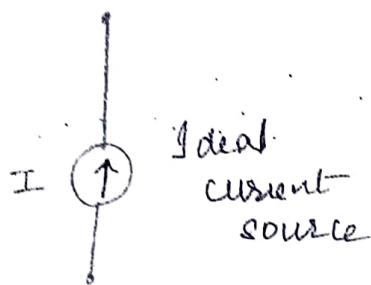
→ Real source can be represented as an ideal current source in parallel with its internal resistance R_{int} .



When load resistance (R_L) is connected across the terminals of the source, the current (I_L) produced by the source divides between R_{int} and R_L . Consequently, the load current is less than it would if the source was ideal.

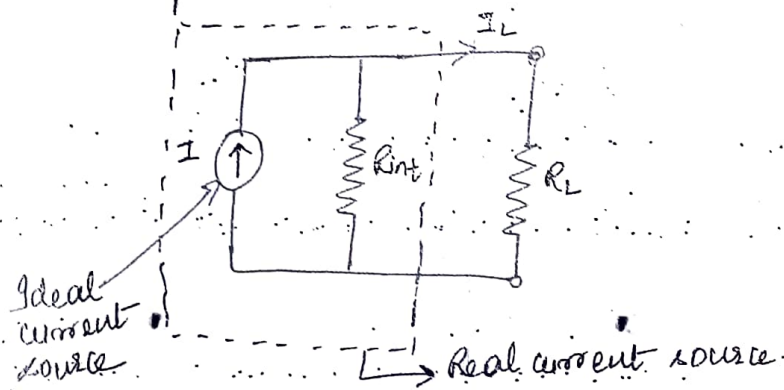
As $R_{int} \uparrow \rightarrow$ current source more closely approaches ideal current source.

Representation:



Practical or Real Current Source:

- A real current source has finite internal resistance. ∴ the load current will change as the value of load resistance R_L changes.
- A real source can be represented as an ideal current source in parallel with its internal resistance (R_{int}).



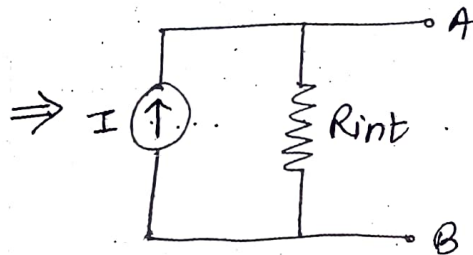
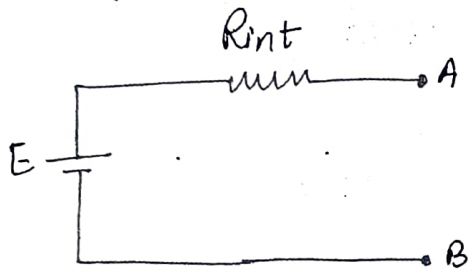
When load resistance (R_L) is connected across the terminals of the source, the current (I) produced by the source divides between R_{int} and R_L .
consequently, the load current is less than it would if the source were ideal.

As $R_{int} \uparrow \uparrow \rightarrow$ current source more closely approaches ideal current source.

Conversion Between Voltage & Current Sources

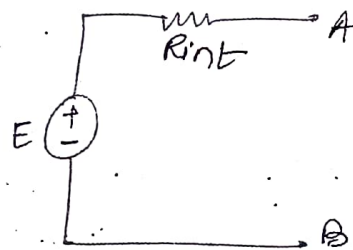
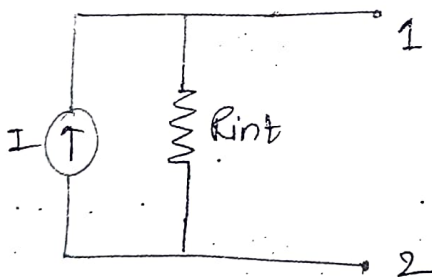
Sources

Voltage Source to Current Source



$$I = \frac{E}{R_{int}}$$

Current to Voltage Source



$$E = I \times R_{int}$$

OHM'S LAW.

Ohm's law states that "the ratio of potential difference (V) between any two points on a conductor to the current (I) flowing between them, is constant, provided the physical conditions (like temperature, pressure, length and cross-sectional area) of the conductor does not change."

$$\frac{V}{I} = \text{constant} = R$$

(or)

"Current flowing through a conductor is directly proportional to the potential difference across the ends of the conductor and inversely proportional to the conductor resistance, provided the physical conditions of the conductor remains unchanged. This relation was discovered by "Georg Simon Ohm" and so it is known as "Ohm's law".

$$I \propto \frac{V}{R}$$

If V = Voltage difference between the two ends of the conductor in Volts (V).

I = Current through the conductor in Ampere.

$$\frac{V}{I} = \text{constant} = R$$

$$I \propto \frac{V}{R} \text{ or } I = \frac{V}{R}$$

$$V = IR$$

where 'R' is called the resistance of the conductor. It is measured in ohms (Ω)

R

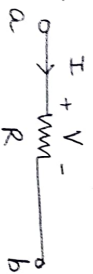
Ohm's law may be expressed in three different forms:

$$V = IR$$

$$I = \frac{V}{R}$$

$$R = \frac{V}{I}$$

Resistance is the electrical property of the material which opposes the current to flow through it.



According to Ohm's law,

$$V_{ab} = IR$$

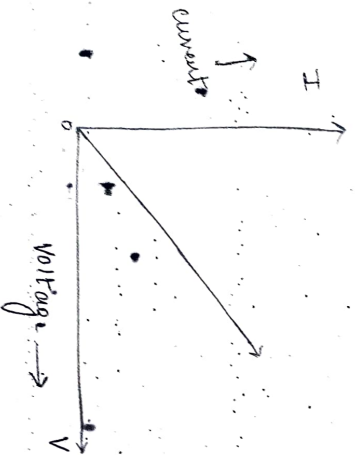
Let V_a = potential at a

V_b = potential at b

Potential across resistor

$$V_{ab} = V_a - V_b$$

Graph



If a curve is plotted between V and I , it is a straight line. The curve between V and I is called the

"V-I characteristic"

$\Rightarrow$ The resistor obey's Ohm's law is called a linear resistor.

$\Rightarrow$ The resistor which does not obey's Ohm's law is called a non-linear resistor.

Conductance :

The reciprocal of resistance i.e. $1/R$ is called the conductance and is denoted by 'G'

It is defined as the inducement offered by the conductor to the flow of current and is measured in siemens (S).

It is also measured in mho (Ω)

1 siemen = 1 mho

$$G = \frac{1}{R}$$

LIMITATION OF OHM'S LAW :

- 1) Ohm's law cannot be applied to circuits consisting of electronic tubes or transistors because such elements are not bilateral i.e. They behave in different ways when the direction of flow of current is reversed as in case of diode.
eg: Zener diodes and voltage regulator (VR) tubes.
- 2) Ohm's law also cannot be applied to circuits consisting of non-linear elements or non-metallic conductors such as powdered carbon, thyrite, electronic arc etc.
eg: for silicon carbide, relationship is given by
$$V = KI^m$$

where, 'K' and 'm' are constants
& 'm' is less than unity.

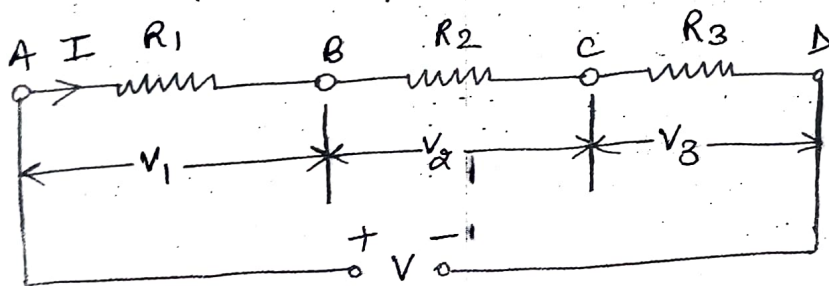
APPLICATIONS OF OHM'S LAW

- Ohm's law can be applied to a complete circuit or to any part of the circuit provided that V, I and R relate to some part.
- When Ohm's law is applied to an entire circuit, values of V, I and R must be used for the entire circuit. When used for certain part of a circuit, values of V, I & R must be used for only that part.

SERIES & PARALLEL COMBINATION OF RESISTANCES

SERIES CIRCUITS:

When the resistors are connected end to end, so that they form only one path for the flow of current, then resistors are said to be connected in series and such circuits are known as series circuits.



Let resistors R_1 , R_2 & R_3 be connected in series, and potential difference of V volts be applied between extreme ends A and D to cause flow of current I amp. through all the resistors R_1 , R_2 & R_3 .

Now according to ohm's law,

$$\begin{aligned} \text{Voltage drop across resistor } R_1, & \quad V_1 = IR_1 \\ \text{Voltage drop across resistor } R_2, & \quad V_2 = IR_2 \\ \text{Voltage drop across resistor } R_3, & \quad V_3 = IR_3 \end{aligned}$$

$$\begin{aligned} \text{Voltage drop across whole circuit,} \\ V = (\text{V.D. across } R_1) + (\text{V.D. across } R_2) + (\text{V.D. across } R_3) \end{aligned}$$

$$\begin{aligned} V &= IR_1 + IR_2 + IR_3 \\ &= I(R_1 + R_2 + R_3) \end{aligned}$$

$$\frac{V}{I} = R_1 + R_2 + R_3$$

$$\frac{V}{I} = R_1 + R_2 + R_3$$

Acc. to Ohm's law $\left(\frac{V}{I} = R \right)$

$$\boxed{R = R_1 + R_2 + R_3} \Rightarrow \text{Effective Resistance of the series circuit}$$

↓
whole circuit resistance.

Thus, when number of resistors are connected in series, the equivalent resistance is given by the arithmetic sum of their individual resistances.

$$\text{i.e. } R = R_1 + R_2 + R_3 + \dots + R_n$$

Important Points:

1. Same current flows through all parts of the circuit.
2. Applied voltage is equal to the sum of voltage drops across the different parts of the circuit.
3. Different resistors have their individual voltage drops.
4. Voltage drop across individual resistor is directly proportional to its resistance, current being the same in each resistor.
5. Voltage drops are additive.
6. Resistances are additive.
7. Powers are additive.

Examples:

- 1) Tube filaments in small radios are in series.
- 2) Fuses are in series with the equipment they protect.
- 3) A thermostat switch is in series with the heating element in an electric iron.
- 4) Rheostats are in series with the coils in large motors for motor current control.

Disadvantages.

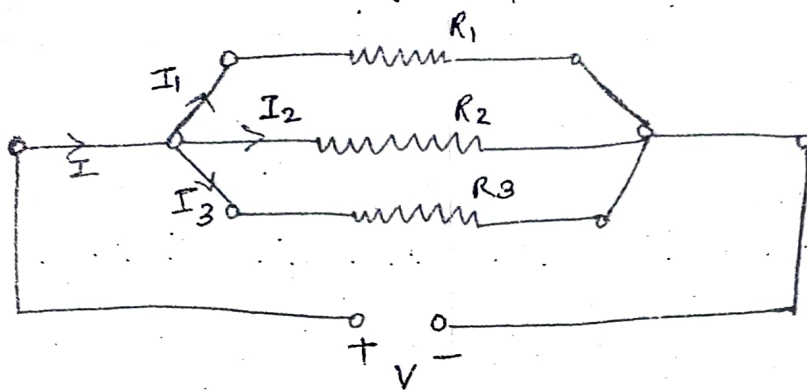
- ① If break occurs in any part of the circuit, no current will flow and the entire circuit become useless.
- ② It is not practicable for lighting circuits.
e.g. each 'R' required 230V.
If three resistors are connected in series,
total supply voltage should be $230 \times 3 = 690 \text{ Volts}$.
If 2 more resistors were added to the ckt.
total voltage = $5 \times 230 = 1150 \text{ Volts}$.
So high supply voltage is required, if no. of ~~resistors~~ resistors increases.
- ③ All devices connected in series must be designed to function at the same current magnitude.
e.g. Electrical devices like heater, toaster, lamp etc. have different current ratings so they cannot be connected in series.

These limitations ~~can~~ are overcome in parallel circuits.

Parallel Circuits:

When a number of resistors are connected in such a way that one end of each of them is joined to a common point and the other ends being joined to another common point, then resistors are said to be connected in parallel and such circuits are known as parallel circuits.

In these circuits current is divided into as many path as the number of resistances.



Let the resistors R_1, R_2 and R_3 be connected in parallel, and potential difference of ' V ' volts be applied across the circuit.

Since potential difference across each resistor is same and equal to potential difference applied to the circuit i.e. ' V '

∴ Acc. to Ohm's law,

Current in Resistor R_1 , $I_1 = \frac{V}{R_1}$ — (1)

Current in Resistor R_2 , $I_2 = \frac{V}{R_2}$ — (2)

Current in Resistor R_3 , $I_3 = \frac{V}{R_3}$ — (3)

Total current I ,

$$I = I_1 + I_2 + I_3$$

$$I = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$I = V \left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right]$$

$$\frac{I}{V} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\boxed{\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

Thus, when number of resistors are connected in parallel, the reciprocal of the equivalent resistance

is given by the arithmetic sum of the reciprocals of their individual resistances.

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$$

$$G = G_1 + G_2 + G_3 + \dots + G_n$$

Important Points:

- 1) Same voltage acts across all branches of the circuit.
- 2) Different resistors (or branches) have their individual currents.
- 3) Total circuit current is equal to the sum of individual currents through the various resistors (or branches).
- 4) Branch currents are additive.
- 5) Conductances are additive.
- 6) Powers are additive.
- 7) The reciprocal of the equivalent resistance is equal to the sum of the reciprocals of the resistances of the individual branches.

eg: Various lamps and appliances in a house are connected in parallel, so that each one can be operated independently.

Advantages:

- 1) If a break occurs in any one of the branch circuits, it will have no effect on other branch circuits.
- 2) The appliances rated for same voltage, but different powers can be connected in parallel without disturbing each other's performance.

$\frac{230V}{230W}$
 TV receiver

in
|||
with

$\frac{230V}{40W}$
 lamp

can operate independently.